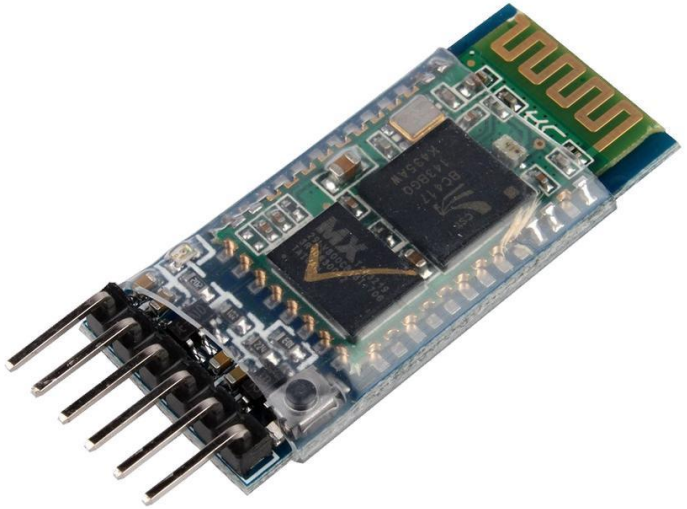
بسم الله الرحمن الرحيم

Bluetooth Module Programming

with Arduino



By

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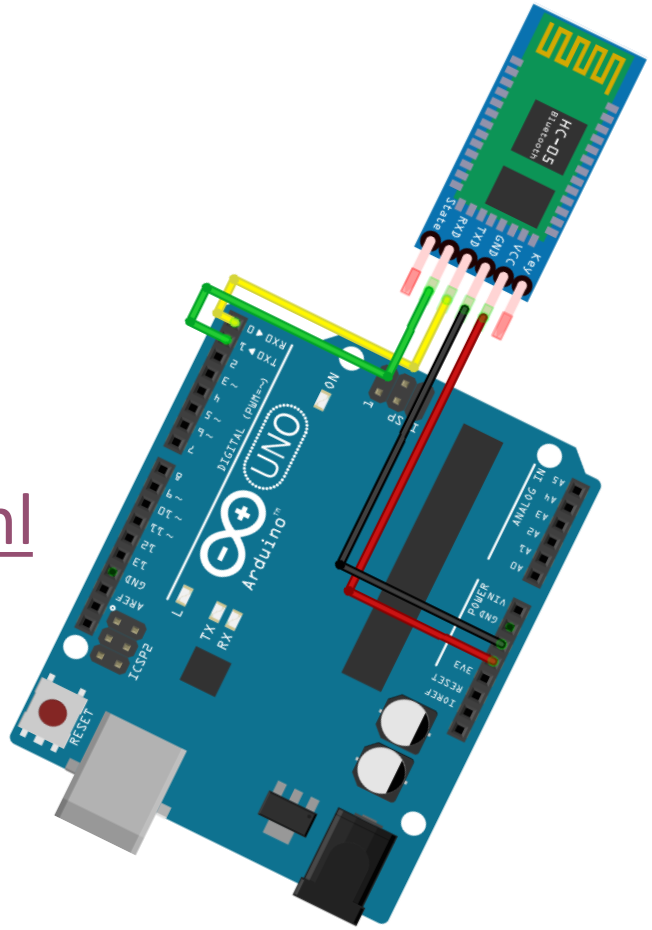
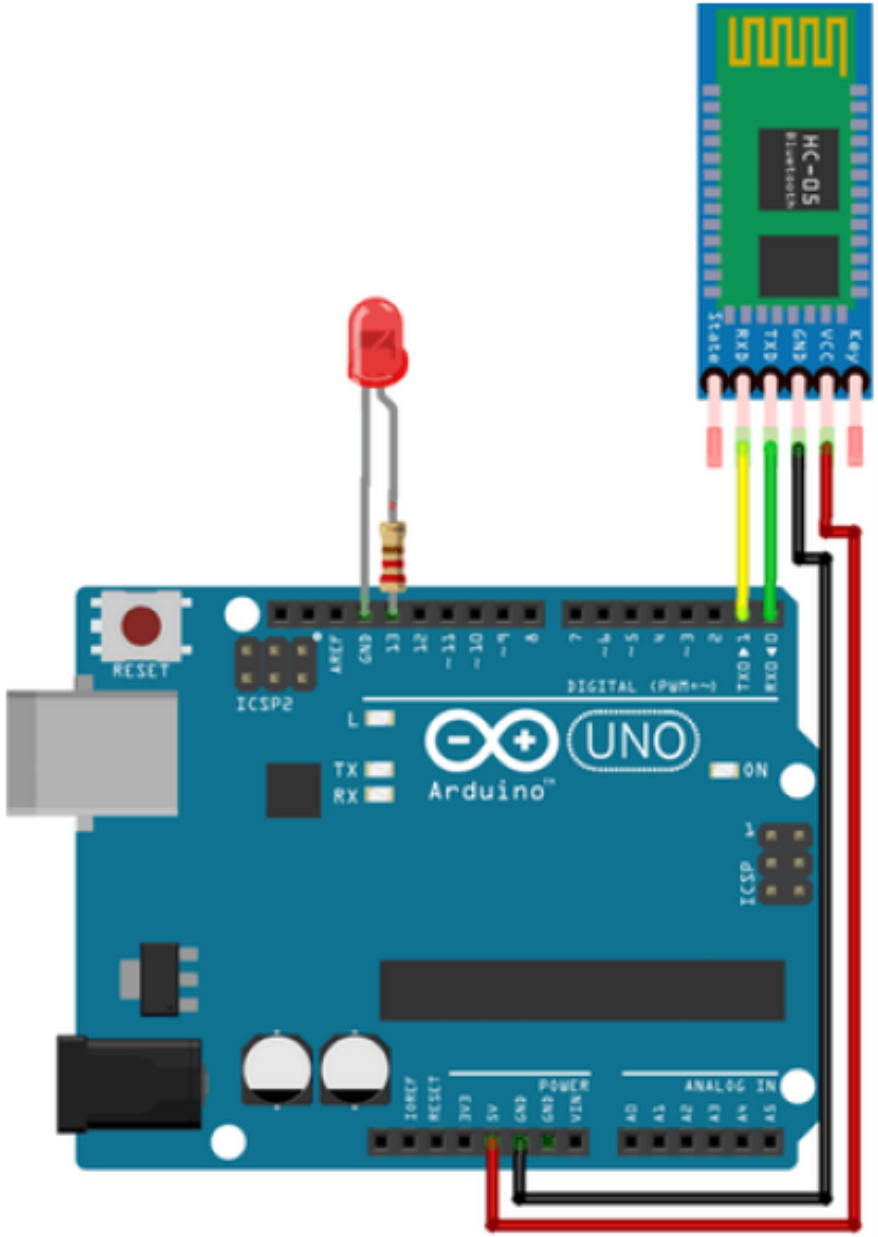


Connect a LED negative to GND of Arduino and positive to pin 13 with a resistance valued between 220Ω – 1KΩ.

Parts Needed

- (1) Arduino Uno
- (1) USB A-to-B Cable
- (1) Breadboard – Half Size
- (1) HC 05 Bluetooth Module
- (1) **Android device (Smart phone or Tablet)**
- (1) LED 5mm
- (1) 220 Ω Resistor
- (4) Jumper Wires

Project Diagram



```
char Incoming_value = 0;            //Variable for storing Incoming_value

void setup()
{
  Serial.begin(9600);            //Sets the data rate in bits per second (baud) for serial data transmission
  pinMode(13, OUTPUT);        //Sets digital pin 13 as output pin
}

void loop()
{
  if(Serial.available() > 0)
  {
    Incoming_value = Serial.read();     //Read the incoming data and store it into variable Incoming_value
    Serial.print(Incoming_value);         //Print Value of Incoming_value in Serial monitor
    Serial.print("\n");                        //New line
    if(Incoming_value == '1')                //Checks whether value of Incoming_value is equal to 1
        digitalWrite(13, HIGH);             //If value is 1 then LED turns ON
    else if(Incoming_value == '0')       //Checks whether value of Incoming_value is equal to 0
        digitalWrite(13, LOW);            //If value is 0 then LED turns OFF
  }
}
```

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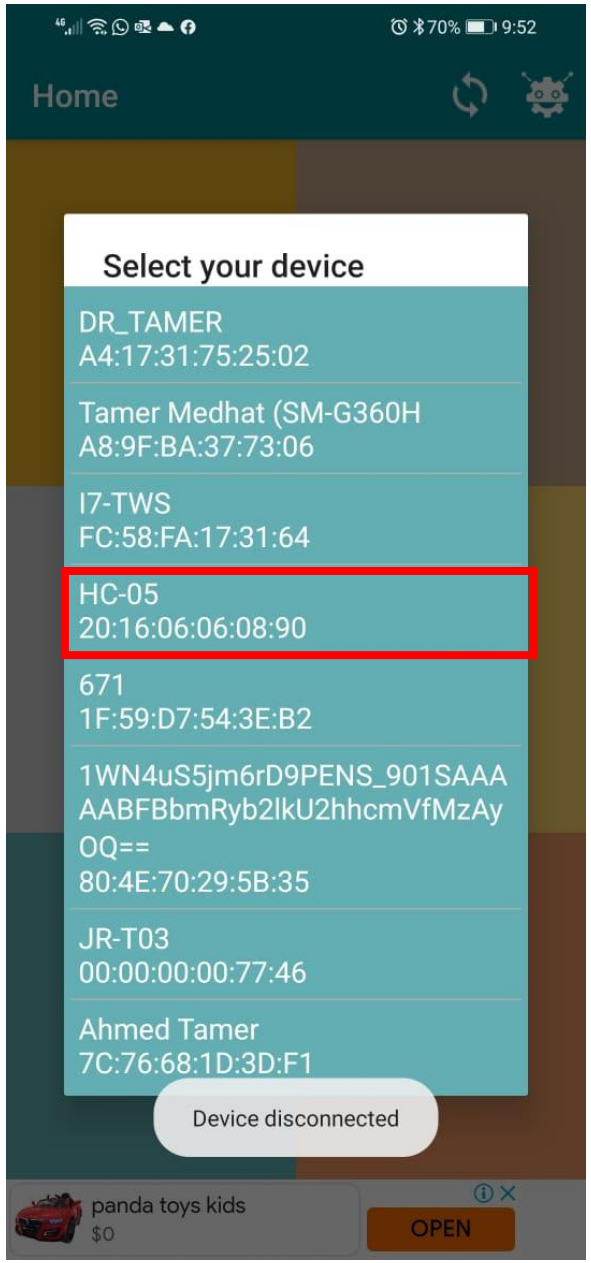
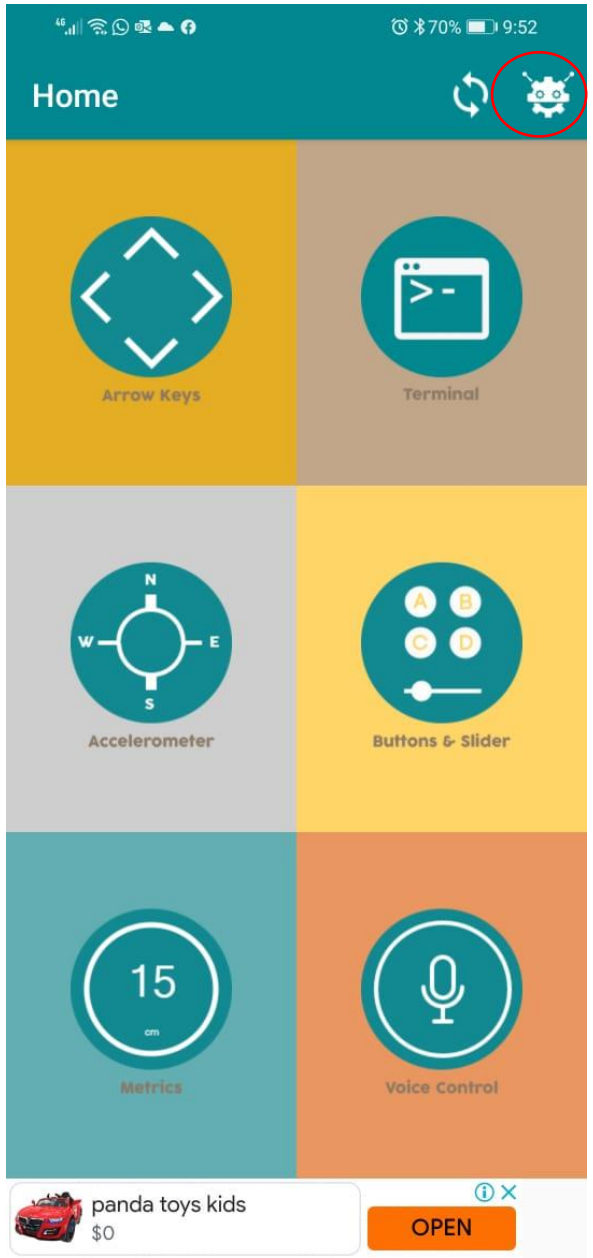
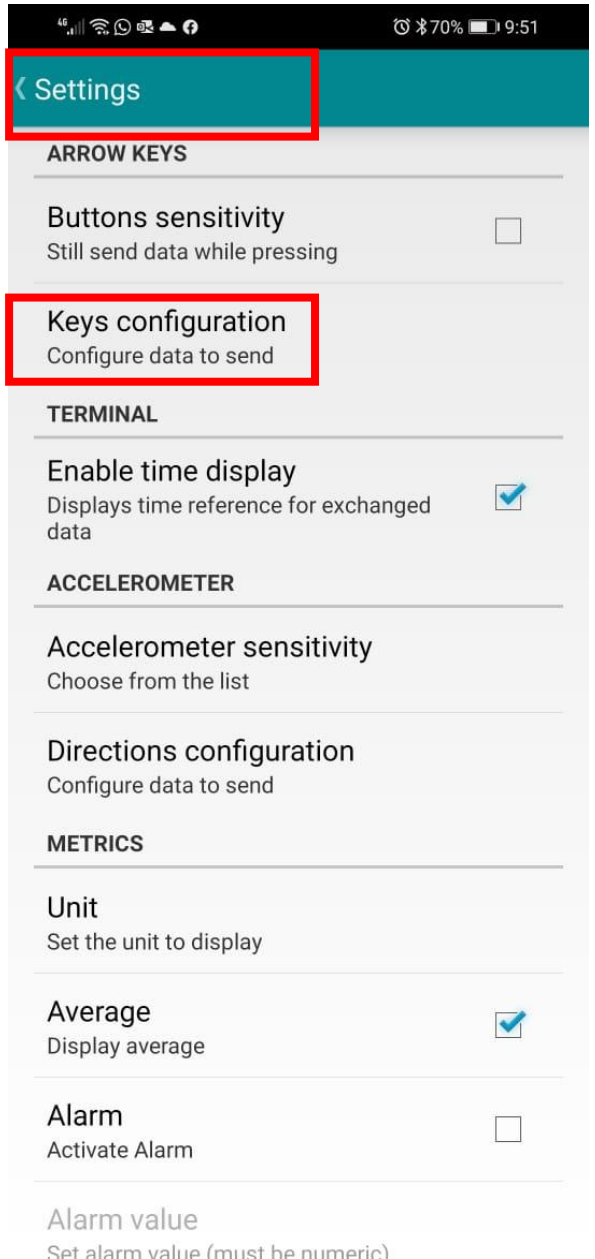
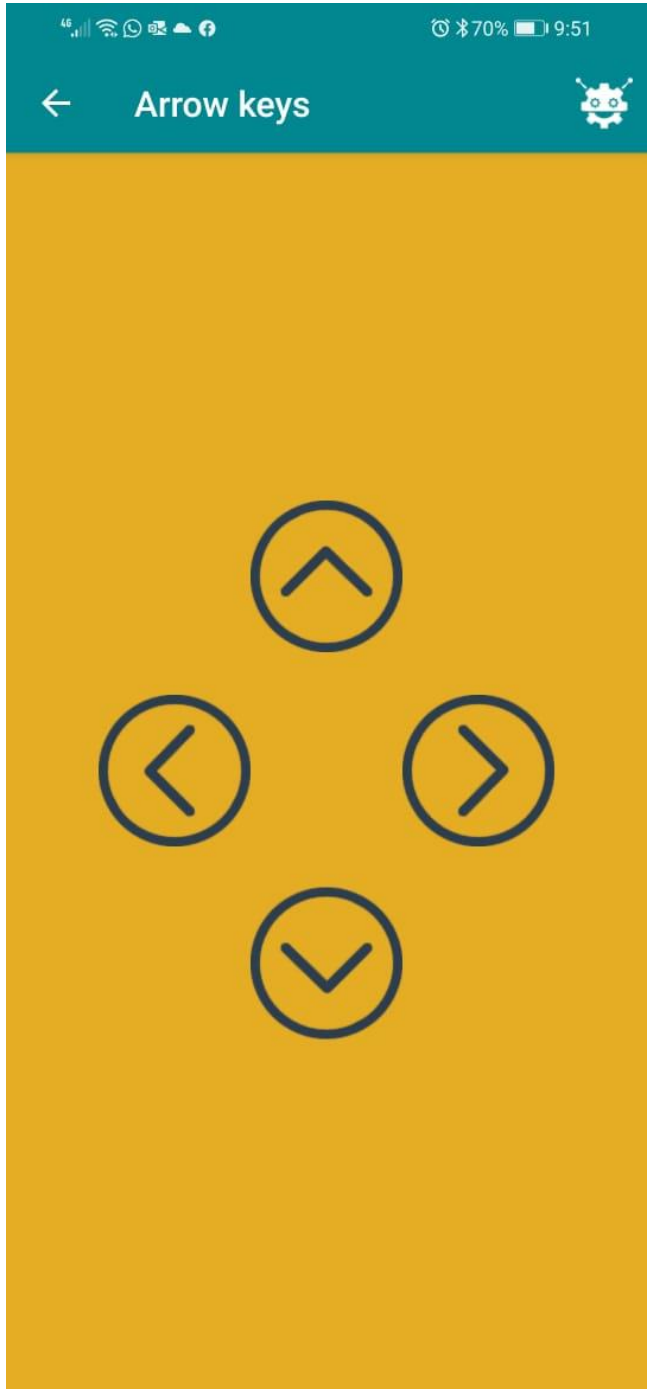
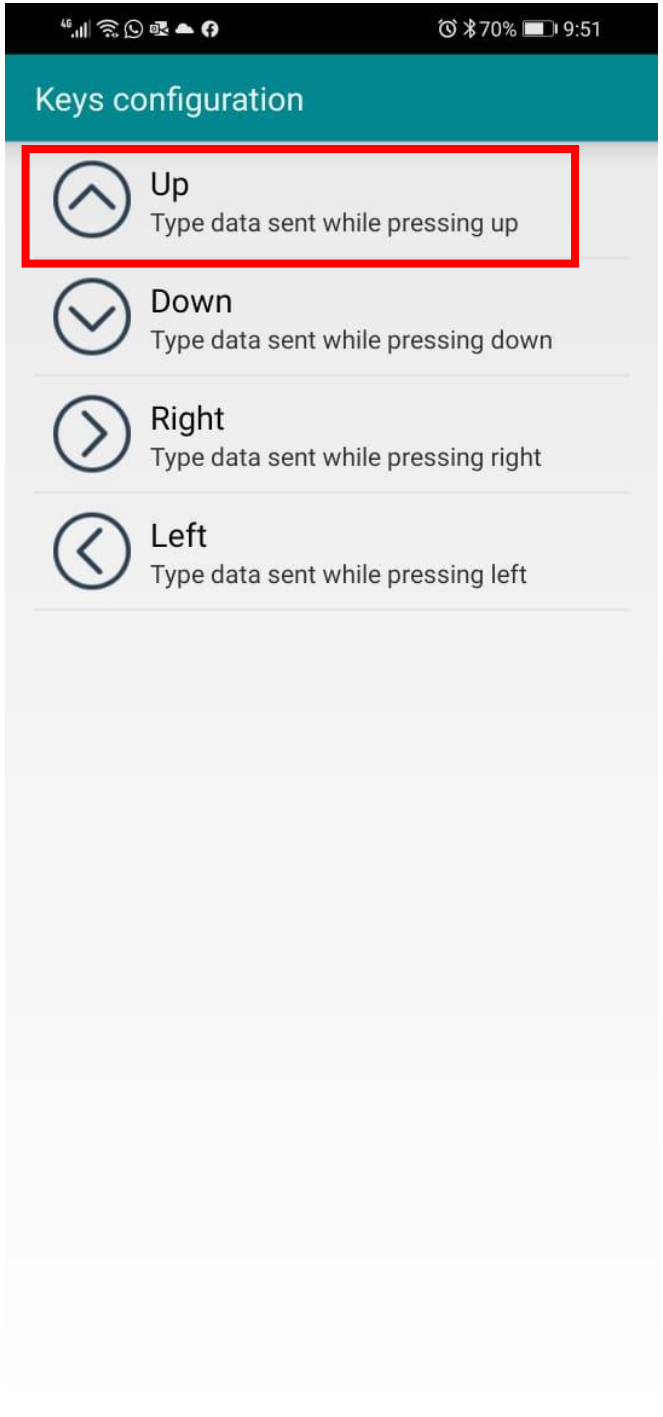
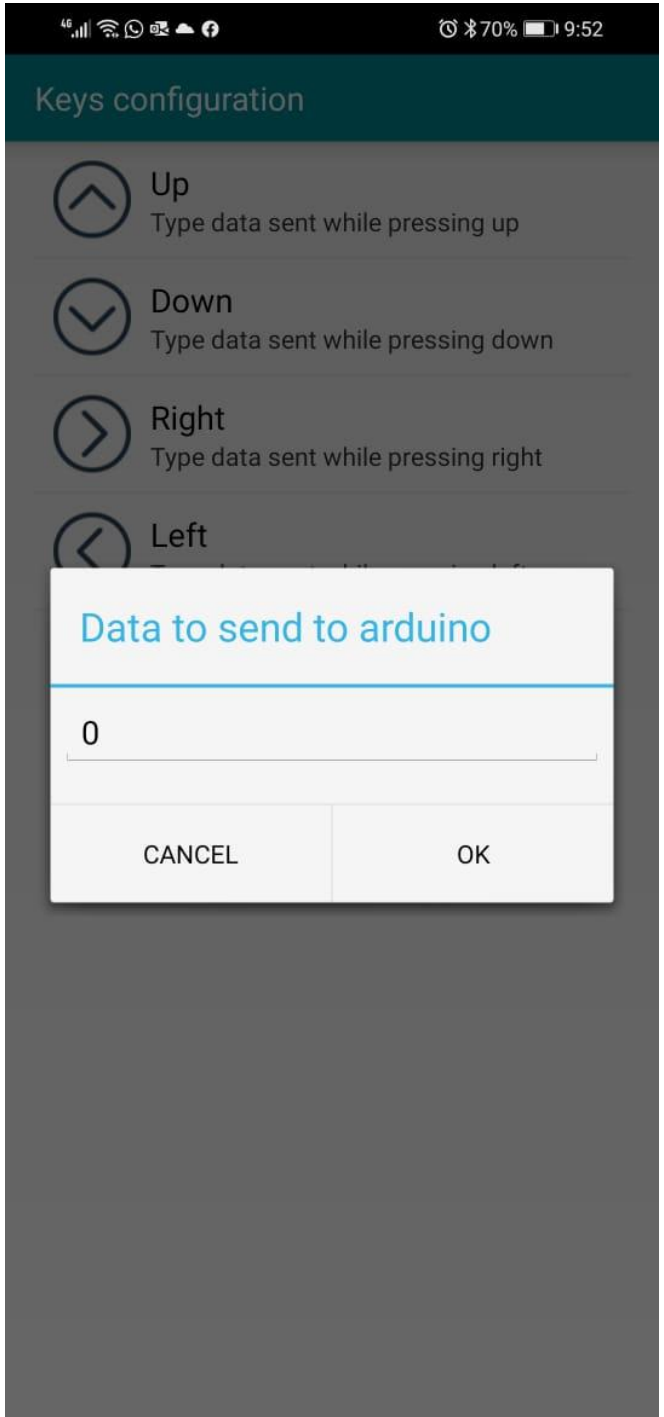


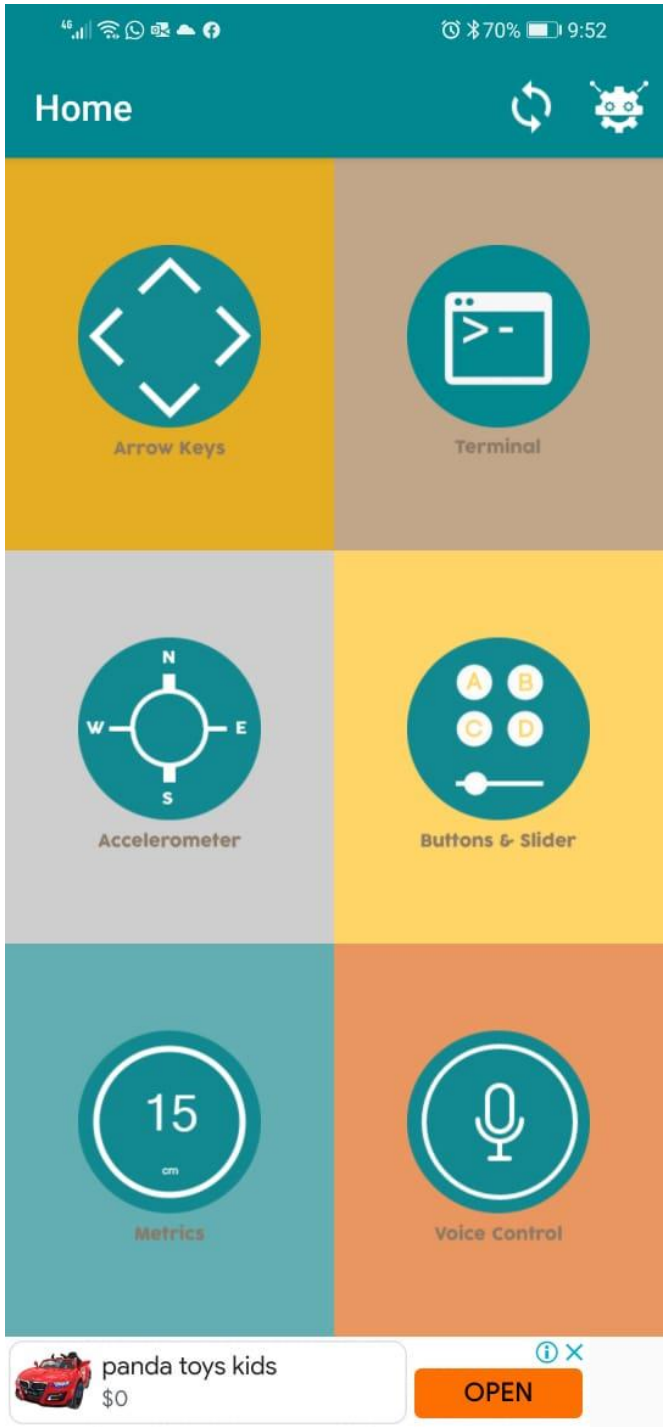


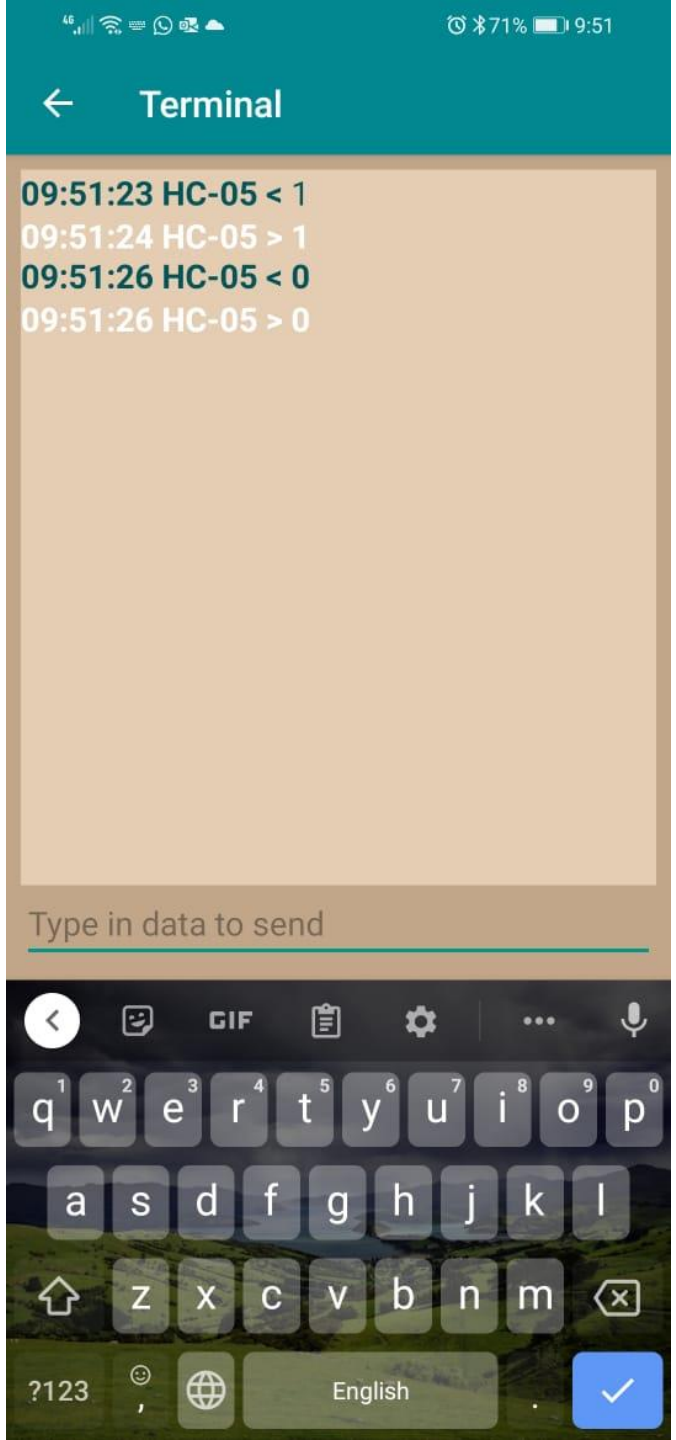



70% 9:51
Arrow keys

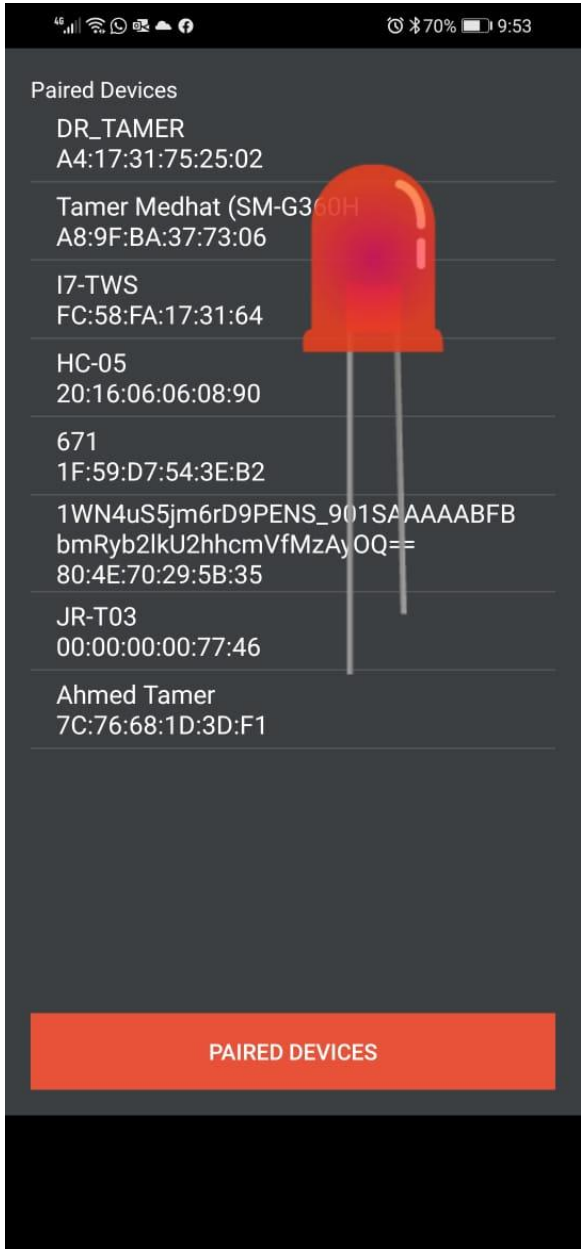

70% 9:51
Keys configuration
Up
Type data sent while pressing up
Down
Type data sent while pressing down
Right
Type data sent while pressing right
Left
Type data sent while pressing left

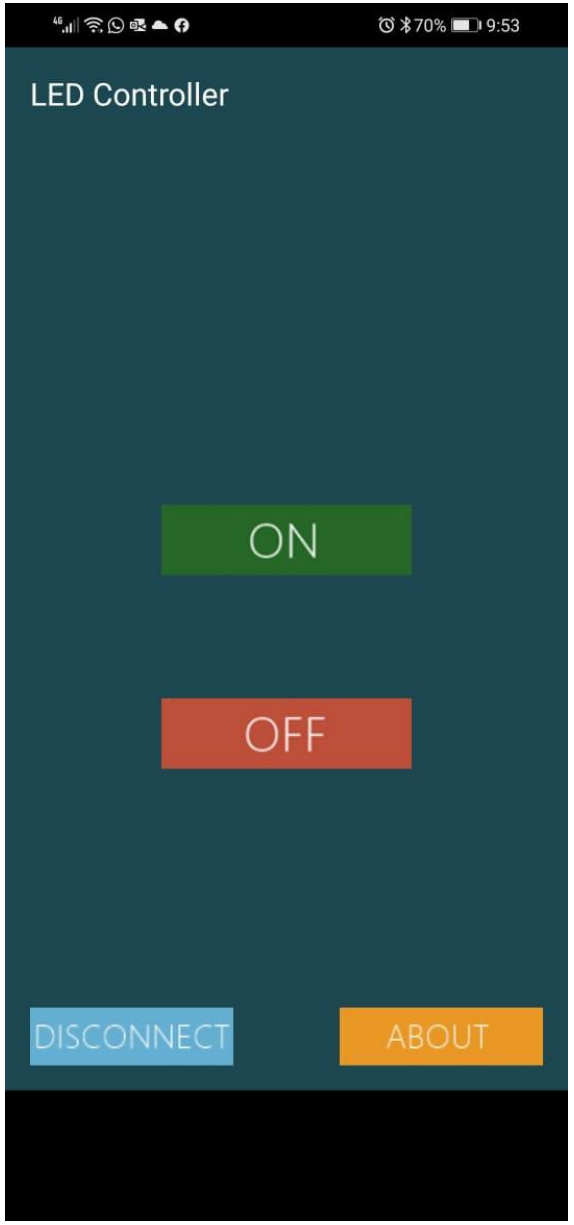

70% 9:52
Keys configuration
Up
Type data sent while pressing up
Down
Type data sent while pressing down
Right
Type data sent while pressing right
Left
Data to send to arduino
0
CANCEL
OK


70% 9:52
Home
Arrow Keys
Terminal
N
W
E
S
Accelerometer
A
B
C
D
Buttons & Slider
15
cm
Metrics
Voice Control
panda toys kids
$0
OPEN


71% 9:51
Terminal
09:51:23 HC-05 < 1
09:51:24 HC-05 > 1
09:51:26 HC-05 < 0
09:51:26 HC-05 > 0
Type in data to send
GIF
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123
English

LED Controller.apk





for

The for statement is used to repeat a block of statements enclosed in curly braces a specified number of times. An increment counter is often used to increment and terminate the loop. There are three parts, separated by semicolons (;), to the for loop header:

```
for (initialization; condition; expression)
{
  doSomething;
}
```

The initialization of a local variable, or increment counter, happens first and only once. Each time through the loop, the following condition is tested. If the condition remains true, the following statements and expression are executed and the condition is tested again. When the condition becomes false, the loop ends.

The following example starts the integer i at 0, tests to see if i is still less than 20 and if true, increments i by 1 and executes the enclosed statements:

```
for (int i=0; i<20; i++)  // declares i, tests if less
{                         // than 20, increments i by 1
  digitalWrite(13, HIGH); // turns pin 13 on
  delay(250);             // pauses for 1/4 second
  digitalWrite(13, LOW);  // turns pin 13 off
  delay(250);             // pauses for 1/4 second
}
```

Note: The C for loop is much more flexible than for loops found in some other computer languages, including BASIC. Any or all of the three header elements may be omitted, although the semicolons are required. Also the statements for initialization, condition, and expression can be any valid C statements with unrelated variables. These types of unusual for statements may provide solutions to some rare programming problems.

Questions ?

Thank you